

Deepika Chopra

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 76% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, I2C/SPI/UART, MQTT, ARM Cortex, FreeRTOS, PCB Debugging, Python

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 45%.
- Led a squad of 3 engineers delivering scalable microservices processing 9M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 65%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2015 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$34k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [C/C++, I2C/SPI/UART, MQTT, ARM Cortex]

Engineered end-to-end production solution leveraging C/C++ and I2C/SPI/UART with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [MQTT, ARM Cortex, FreeRTOS, PCB Debugging]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

NIT Trichy — 2011 - 2015 (Score: 9.12 / 10.0)